

Physikalisches Anfängerpraktikum der Universität Heidelberg

Versuch 12

Moment of Inertia

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1 Introduction

1.1 AI Usage

AI was used only in the Introduction to help with formulation.

1.2 Goal

The goal of this experiment is to investigate the damping of a rotary pendulum using different measurement methods. The damping constant δ is determined from the decrease of the amplitude of a free oscillation, the half-width of a resonance curve, and the resonance amplification. The results are then compared.

1.3 Foundations

The experiment uses a rotary pendulum (Pohl's wheel). For a damped free oscillation, the angular displacement is

$$\varphi(t) = \varphi_0 e^{-\delta t} \sin(\omega_f t), \quad (1)$$

where δ is the damping constant and ω_f is the angular frequency of the damped oscillation. The amplitude therefore decreases exponentially:

$$a(t) = a_0 e^{-\delta t}. \quad (2)$$

The damping constant can therefore be determined from the half-life of the amplitude, defined by $a(t_{\frac{1}{2}}) = \frac{a_0}{2}$:

$$\delta = \frac{\ln(2)}{t_{\frac{1}{2}}}. \quad (3)$$

The damped and undamped angular frequencies are related by

$$\omega_f = \sqrt{\omega_0^2 - \delta^2}. \quad (4)$$

When the pendulum is driven periodically with angular frequency ω , a forced oscillation is produced. After the transient motion has decayed, the stationary amplitude is

$$b(\omega) = \frac{A\omega_0^2}{\sqrt{(\omega_0^2 - \omega^2)^2 + (2\delta\omega)^2}}, \quad (5)$$

where A is the amplitude of the excitation. The amplitude reaches its maximum at the resonance frequency

$$\omega' = \sqrt{\omega_0^2 - 2\delta^2}. \quad (6)$$

For weak damping, $\omega' \approx \omega_0$. The damping can also be obtained from the half-width H of the resonance curve:

$$H = \omega_2 - \omega_1 = 2\delta, \quad (7)$$

where ω_1 and ω_2 are the frequencies at which the squared amplitude has fallen to half its maximum value. Another measure of damping is given by the resonance amplification,

$$\frac{b(\omega')}{b(\omega \rightarrow 0)} = \frac{\omega_0}{2\delta}. \quad (8)$$

1.4 Measurement

The experiment is carried out using a rotary pendulum with an eddy-current brake for controlled damping. A stepper motor with an eccentric provides the periodic excitation. Its controller and power supply drive the motor, while a function generator provides the TTL square-wave control signal. A separate power supply is used for the eddy-current brake. The setup is shown in the sketch below.

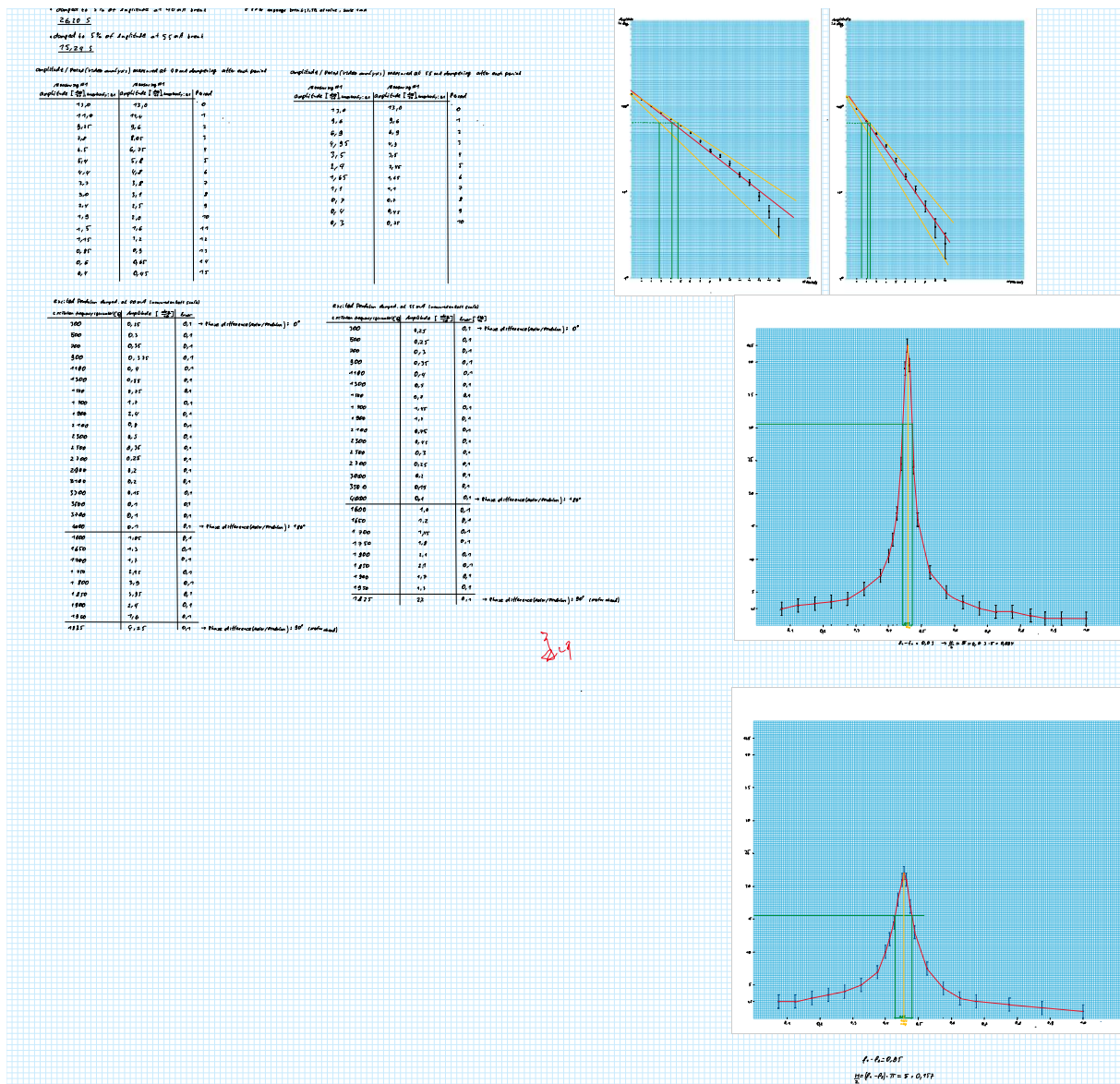
First, the undamped oscillation period T_0 is measured by timing three times 20 oscillations.

The damping is then switched on for two different brake currents. For each current, the time it takes for the amplitude to decay to 5% of the original amplitude is recorded and is used to determine how long to perform the following experiment:

The amplitude of the freely oscillating pendulum is recorded at successive reversal points and plotted on a logarithmic scale.

For the forced oscillation, the stationary amplitude is measured for different excitation frequencies between 0.1 and 1Hz and for both damping values. The resonance curve is used to determine the resonance frequency, half-width, and resonance amplification. The phase difference between the exciter and the pendulum is also observed at low frequency, near resonance, and at high frequency.

2 Protocol



5

2.1 Sketch

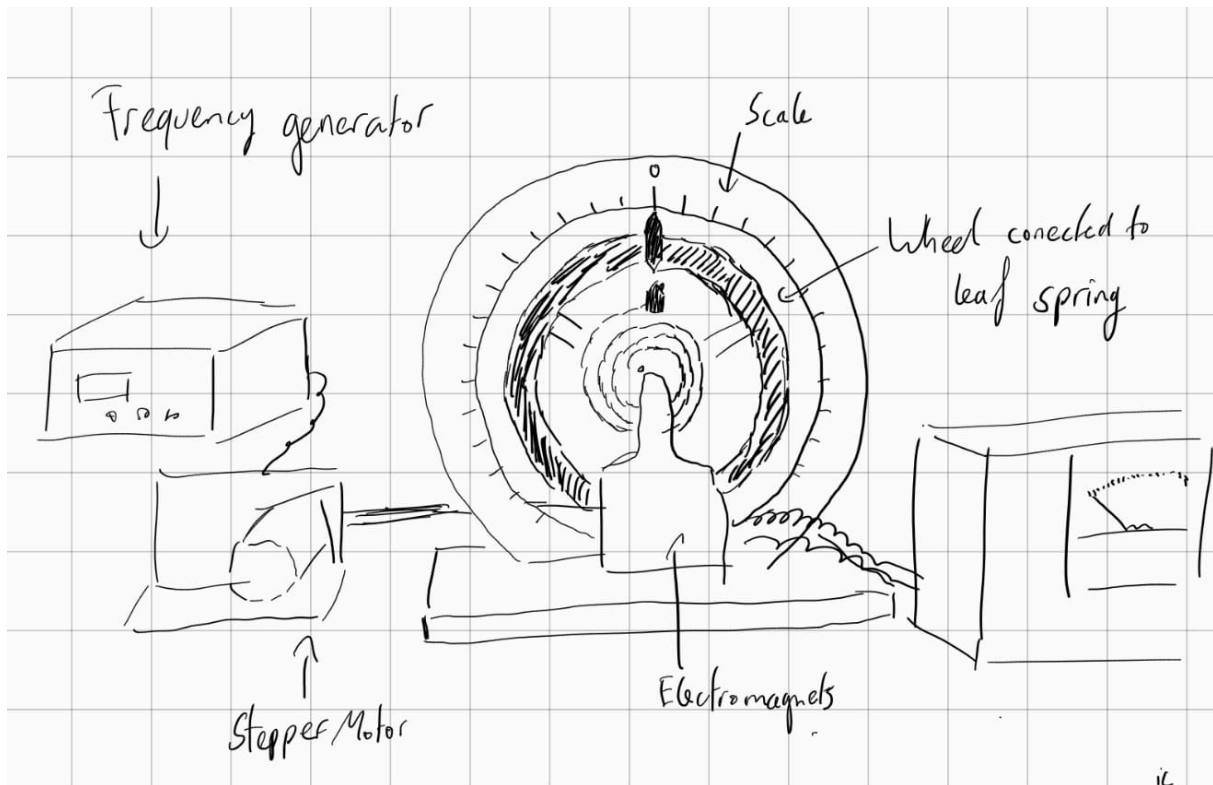


Abbildung 2: Sketch of the Setup

3 Evaluation

3.1 Determining the oscillation period

We measured 3 times 20 oscillations using a stopwatch. We assume the error due to the reaction time to be about 200 ms (we neglect the much smaller internal error of the Stopwatch)

As the first step, we divided all measured times by 20 and calculated the average error of a measurement S_E .

We added the error of the mean value $S_M = \frac{S_E}{\sqrt{3}}$ to the reaction time according to the laws of error propagation:

$$\Delta T = \sqrt{S_M^2 + \left(\frac{200 \text{ ms}}{20}\right)^2} \approx 0.010 \quad (9)$$

This results in a Period of

$T = (2.319 \pm 0.010) \text{ s}$	(10)
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In practice, we used the python package `uncertainties` to keep track of the errors, but I am going to write out the important steps here as well.

3.2 Determining the damping constant with the amplitude decrease of the free oscillation

We assumed our error when reading out the amplitude to be $\Delta a = 1^\circ$, because the Resolution of our Scale was 2° .

Abbildung 3 shows the amplitude as function of the number of oscillations. To determine the damping constant, we performed the following steps:

1. matching regression lines through the values
2. determining the half-life of the amplitude $t_{\frac{1}{2}}$

We know from Gleichung 3, that we can compute δ using

$$\delta = \frac{\ln(2)}{T_{\frac{1}{2}}} \quad (11)$$

where $T_{\frac{1}{2}}$ is the half-life of the amplitude in seconds. (We are going to denote the half-life of the amplitude measured in the number of Periods as $t_{\frac{1}{2}}$)

To estimate the error $\Delta\delta$, I estimated the orange lines to be a lower and upper bound on the 1σ -Intervall of the slope of the regression lines. I determined the half-life of these error-lines as well, which gave me a upper and lower bound on the 1σ -Intervall of $t_{\frac{1}{2}}$. I then used the bound which was the farthest from the value of $t_{\frac{1}{2}}$ as an upper bound on the error of $t_{\frac{1}{2}}$, which gave me the following two values for $t_{\frac{1}{2}}$

$$40\text{mA Damping: } t_{\frac{1}{2}} = (3.4 \pm 0.5) \text{ Periods} \quad (12)$$

$$55\text{mA Damping: } t_{1_2} = (1.95 \pm 0.2) \text{ Periods} \quad (13)$$

We have to multiply by the period T that we determined earlier:

$$40\text{mA Damping: } T_{\frac{1}{2}} = t_{\frac{1}{2}} \cdot T = (7.4 \pm 1.1) \text{ s} \quad (14)$$

$$55\text{mA Damping: } T_{1_2} = t_{\frac{1}{2}} \cdot T = (4.2 \pm 0.4) \text{ s} \quad (15)$$

Using Gleichung 11, we get

$$40\text{mA Damping: } \delta = (0.094 \pm 0.014) \text{ s}^{-1} \quad (16)$$

$$55\text{mA Damping: } \delta = (0.163 \pm 0.017) \text{ s}^{-1} \quad (17)$$

Error Propagation by Hand:

$$dT_{\frac{1}{2}} = t_{\frac{1}{2}} dT + T dt_{\frac{1}{2}} \quad (18)$$

$$\Delta T_{\frac{1}{2}} = \sqrt{\left(t_{\frac{1}{2}} \Delta T\right)^2 + \left(T \Delta t_{\frac{1}{2}}\right)^2} \quad (19)$$

$$d\delta = -\frac{\ln(2)}{T_{\frac{1}{2}}^2} dT_{\frac{1}{2}} \quad (20)$$

$$\Delta\delta = \left| -\frac{\ln(2)}{T_{\frac{1}{2}}^2} \Delta T_{\frac{1}{2}} \right| = \frac{\ln(2)}{T_{\frac{1}{2}}^2} dT_{\frac{1}{2}} \quad (21)$$

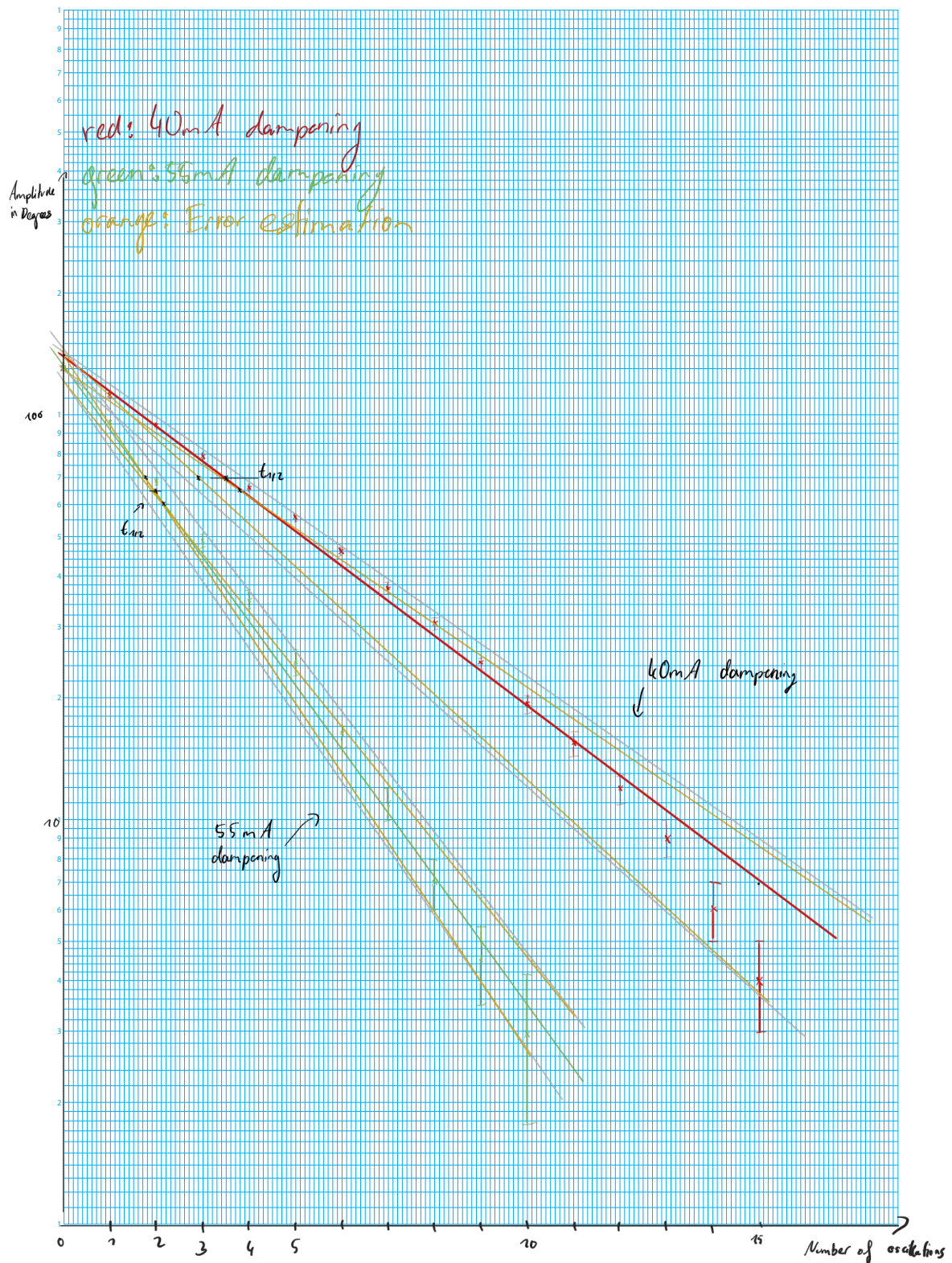


Abbildung 3: Amplitude as a function of the number of oscillations in a logarithmic plot

3.3 Determining the damping constant with a forced oscillation

In Abbildung 4 I plotted the amplitude as a function of the excitation frequency.

We assumed an error of $\Delta a = 1^\circ$ again when reading out the amplitude. We observed the frequency of the frequency generator varying in a range of about $\Delta f_g = 2$ Hz. The error of the scale of the device (0.1Hz) is negligible compared to this.

We also chose to ignore the error of the frequency generator Δf_g varying in a 2 Hz range, because it is negligible compared to the error Δa we had when measuring the amplitude.

3.3.1 Determining the resonance frequency

We read out the resonance frequency from Abbildung 4 assuming that the curve is mostly symmetric in the upper part:

$$40\text{mA Damping: } f_g = (1820 \pm 20) \text{ Hz} \quad (22)$$

$$55\text{mA Damping: } f_g = (1830 \pm 20) \text{ Hz} \quad (23)$$

We get the real resonance frequency of the Pendulum by dividing by 4000 and multiplying by 2π :

$$40\text{mA Damping: } \omega' = (2.859 \pm 0.032) \text{ Hz} \quad (24)$$

$$55\text{mA Damping: } \omega' = (2.875 \pm 0.032) \text{ Hz} \quad (25)$$

One important parameter of the curve is the half-width H , which specifies the difference of the two Frequencies $w_2 - w_1$ where the Amplitude $b(\omega)$ reaches $\frac{b(\omega')}{\sqrt{2}}$ (with the resonance frequency ω').

$$40\text{mA Damping: } H_g = (120 \pm 10) \text{ Hz} \quad (26)$$

$$55\text{mA Damping: } H_g = (200 \pm 10) \text{ Hz} \quad (27)$$

We assumed the error ΔH to be $\Delta H = 10$ Hz because of the resolution of the graph paper (which is 20 Hz).

H_g is still in terms of the generator frequency, we have to divide by 4000 to get H in terms of the excitation frequency:

$$40\text{mA Damping: } H = (0.030 \pm 0.0025) \text{ Hz} \quad (28)$$

$$55\text{mA Damping: } H = (0.050 \pm 0.0025) \text{ Hz} \quad (29)$$

Now we can use Gleichung 7 to calculate δ :

$$40\text{mA Damping: } \delta = (0.094 \pm 0.008) \text{ Hz} \quad (30)$$

$$55\text{mA Damping: } \delta = (0.157 \pm 0.008) \text{ Hz} \quad (31)$$

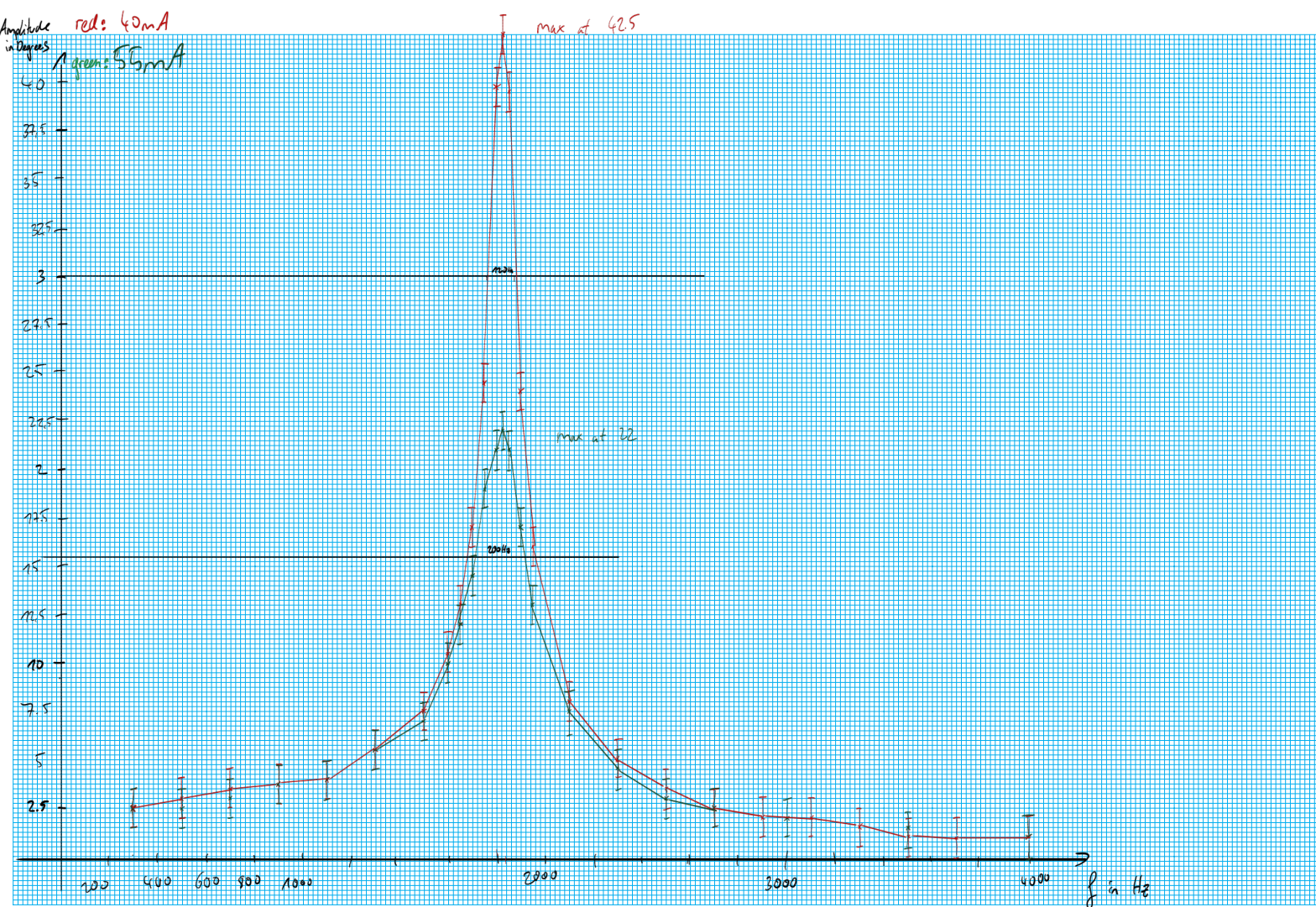


Abbildung 4: Amplitude as a function of the frequency of the motor

3.4 Calculation using resonance amplification

The resonance amplification is given by the ratio

$$\frac{b_{\max}}{b_0} = \frac{\omega_0}{2\delta} \quad (32)$$

with $b_{\max}$ being the maximum amplitude and b_0 being the amplitude as the frequency tends to zero.

Using our measurements for $b_{\max}$, b_0 and $\omega_0 = \frac{2\pi}{T} = (2.884 \pm 0.015) \text{ Hz}$, we can calculate δ :

$$\delta = \frac{\omega_0 b_0}{2b_{\max}} \quad (33)$$

	$b_{\max}$ in °	b_0 in °	δ in s^{-1}
40mA Damping	(42.5 ± 1.0)	(2.5 ± 1.0)	(0.085 ± 0.034)
50mA Damping	(22.0 ± 1.0)	(2.5 ± 1.0)	(0.16 ± 0.07)

Error Propagation by Hand:

$$d\delta = \frac{b_0}{2b_{\max}} dw_0 + \frac{\omega_0}{2b_{\max}} db_0 - \frac{2w_0 b_0}{(2b_{\max})^2} db_{\max} \quad (34)$$

$$\Delta\delta = \sqrt{\left(\frac{b_0}{2b_{\max}} \Delta w_0\right)^2 + \left(\frac{\omega_0}{2b_{\max}} \Delta b_0\right)^2 + \left(\frac{2w_0 b_0}{(2b_{\max})^2} \Delta b_{\max}\right)^2} \quad (35)$$

3.5 Phase shift

For very small frequencies, we observed no phase difference between the exciter and the pendulum. At the resonance frequency, there was a phase shift of $\frac{\pi}{2}$ and at very high frequencies, there was a phase difference of π .

4 Results

$T = (2.319 \pm 0.010) \text{ s}$	(36)
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Method	40 mA damping	55 mA Damping
Amplitude decrease	(0.094 ± 0.014)	(0.163 ± 0.017)
Half-Width	(0.094 ± 0.008)	(0.157 ± 0.008)
Resonance Amplification	(0.085 ± 0.034)	(0.16 ± 0.07)

Tabelle 1: Our values of δ (in Hz)

	Frequency	Standardized mean difference to ω_0
Natural Frequency ω_0	$(2.884 \pm 0.015) \text{ Hz}$	0
40mA Dampening ω'	$(2.859 \pm 0.032) \text{ Hz}$	0.74
55mA Dampening ω'	$(2.875 \pm 0.032) \text{ Hz}$	0.27

Tabelle 2: Resonance Frequency

5 Discussion

To check if the different measurements for δ are within the error margin of each other, i computed the standardized mean difference β between all value pairs:

$$\beta = \frac{|\delta_1 - \delta_2|}{\sqrt{\sigma_1^2 + \sigma_2^2}} \quad (37)$$

40mA Damping	Amplitude decrease	Half-Width	Resonance Amplification
Amplitude decrease	0.0	0.0	0.24
Half-Width	0.0	0.0	0.26
Resonance Amplification	0.24	0.26	0.0

Tabelle 3: β for 40mA Damping

55mA Damping	Amplitude decrease	Half-Width	Resonance Amplification
Amplitude decrease	0.0	0.32	0.04
Half-Width	0.32	0.0	0.04
Resonance Amplification	0.04	0.04	0.0

Tabelle 4: β for 55mA Damping

We can see, that all values are a lot smaller than 1, which means that all values lie well inside 1σ of each other.

5.1 Amplitude decrease

In Abbildung 3, we can observe that the amplitude of the last few oscillations doesn't match the regression line very well.

A possible cause for this could be, that there were other friction factors, that were not proportional to $\dot{\varphi}$. One example might be the turbulent air friction which is proportional to $\dot{\varphi}^2$. For the very small velocities, there is also nonlinear friction effects at the transition from sliding friction to static friction (stick-slip). This could be the case inside the bearing of the Pendulum when it is swinging at very small amplitudes where it spends more time at very slow velocities.

This method would be a lot more accurate for Experiments that run for a lot more Periods (like the Pendulum in Experiment 14).

5.2 Half Width

Estimating the error of the Half-width that we measured graphically proved to be quite difficult. Theoretically, we would have to take into account the error of the resonanc-curve that we fitted to our measurement, but this is difficult to estimate, especially on paper. I approximated the curve by just connecting our measurement with straight lines. This is reasonably accurate, because we took many measurements in the area aorund the peak, where the slope changes a lot.

This Method could be even more accurate with more measurements around the Peak and the half-width. Using an algorithm to fit a curve to the points could also help with the accuracy and be useful to get a proper estimate of the error.

5.3 Resonance Amplification

The error of the resonance amplification is very big compared to the other methods. The main reason for this, is that we had an error of $\Delta a = 1^\circ$ for measuring the amplitude, which is very big compared to the small amplitudes ($b_0 = 2.5^\circ$).

Using a motor with a bigger crank, could make this Method a lot more accurate by increasing the amplitude for very small excitation frequencies. This could run into the issue of the Spring not behaving linearly anymore for very big amplitudes at the resonance frequency.

5.4 Resonance Frequency

We can't observe anything useful about the resonance frequencies for the different amounts of damping, because we can't determine them precisely enough with this experiment. As you can see in Tabelle 2, they are all inside of less than 1σ of the natural Frequency.

To get better results, the Period would have to be determined more precisely to compute the natural frequency with less error. A higher density of amplitude measurements around the resonance frequency would also increase the precision of this experiment more.